Question 1: What is a Decision Tree, and how does it work in the context of classification?

A Decision Tree is a supervised machine learning algorithm used for classification and regression tasks. It represents decisions in the form of a tree-like structure consisting of a root node, internal nodes, branches, and leaf nodes.

In classification, the algorithm starts at the root node and repeatedly splits the dataset based on feature values. Each split is chosen to best separate the classes. The process continues until a stopping condition is reached. The final leaf node represents the predicted class label. Decision Trees are easy to understand, interpret, and visualize.

Question 2: Explain the concepts of Gini Impurity and Entropy as impurity measures. How do they impact the splits in a Decision Tree?

Gini Impurity measures the probability of incorrectly classifying a randomly chosen sample if it were labeled according to the class distribution in a node. A lower Gini value indicates a purer node.

Entropy measures the amount of uncertainty or disorder in a node. Lower entropy means the node contains mostly samples from a single class.

Both measures help determine the best split in a Decision Tree. The algorithm selects the split that produces the greatest reduction in impurity, resulting in more homogeneous child nodes and improved classification performance.

Question 3: What is the difference between Pre-Pruning and Post-Pruning in Decision Trees? Give one practical advantage of using each.

Pre-Pruning stops the growth of the tree before it becomes too complex by applying constraints such as maximum depth or minimum samples per node.

Post-Pruning allows the tree to grow fully and then removes branches that do not significantly improve performance.

Advantage of Pre-Pruning: Reduces training time and computational cost.

Advantage of Post-Pruning: Often produces better accuracy because it considers the complete tree structure before removing unnecessary branches.

Question 4: What is Information Gain in Decision Trees, and why is it important for choosing the best split?

Information Gain is a metric that measures the reduction in entropy after splitting a dataset on a particular feature. It quantifies how much useful information a split provides.

A higher Information Gain indicates a better split because it creates purer subsets. Decision Tree algorithms use Information Gain to select the feature that best separates the data at each node, leading to more accurate predictions.

Question 5: What are some common real-world applications of Decision Trees, and what are their main advantages and limitations?

Applications:

- Medical diagnosis
- Credit risk assessment
- Fraud detection
- Customer churn prediction
- Loan approval systems
- Marketing and customer segmentation

Advantages:

- Easy to understand and interpret
- Requires little data preprocessing
- Can handle both numerical and categorical data
- Supports classification and regression tasks

Limitations:

- Prone to overfitting
- Sensitive to small changes in data
- May not perform as well as ensemble methods on complex datasets
- Can create biased trees when classes are imbalanced